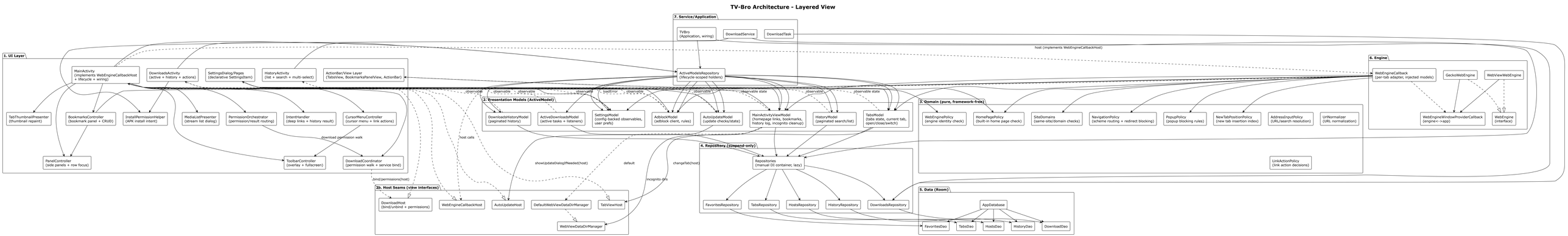
TV-Bro Architecture – Layered View
7. Service/Application
TVBro
(Application, wiring)
DownloadService
DownloadTask
host (implements WebEngineCallbackHost)
1. UI Layer
MainActivity
(implements WebEngineCallbackHost
+ lifecycle + wiring)
DownloadsActivity
(active + history + actions)
SettingsDialog/Pages
(declarative SettingsItem)
HistoryActivity
(list + search + multi-select)
ActionBar/View Layer
(TabsView, BookmarksPanelView, ActionBar)
ActiveModelsRepository
(lifecycle-scoped holders)
6. Engine
WebEngineCallback
(per-tab adapter, injected models)
GeckoWebEngine
WebViewWebEngine
WebEngineWindowProviderCallback
(engine<->app)
WebEngine
(interface)
observe
observable
badError
observable state
TabThumbnailPresenter
(thumbnail repaint)
BookmarksController
(bookmark panel + CRUD)
InstallPermissionHelper
(APK install intent)
MediaListPresenter
(stream list dialog)
PermissionOrchestrator
(permission/result routing)
IntentHandler
(deep links + history result)
CursorMenuController
(cursor menu + link actions)
download permission walk
PanelController
(side panels + row focus)
ToolbarController
(overlay + fullscreen)
DownloadCoordinator
(permission walk + service bind)
2. Presentation Models (ActiveModel)
DownloadsHistoryModel
(paginated history)
ActiveDownloadsModel
(active tasks + listeners)
SettingsModel
(config-backed observables, user prefs)
AdblockModel
(adblock client, rules)
AutoUpdateModel
(update checks/state)
MainActivityViewModel
(homepage links, bookmarks, history log, incognito cleanup)
HistoryModel
(paginated search/list)
TabsModel
(tabs state, current tab, open/close/switch)
3. Domain (pure, framework-free)
WebEnginePolicy
(engine identity check)
HomePagePolicy
(built-in home page check)
SiteDomains
(same-site/domain checks)
NavigationPolicy
(scheme routing + redirect blocking)
PopupPolicy
(popup blocking rules)
NewTabPositionPolicy
(new tab insertion index)
AddressInputPolicy
(URL/search resolution)
UrlNormalizer
(URL normalization)
LinkActionPolicy
(link action decisions)
host calls
showUpdateDialogIfNeeded(host)
default
changeTab(host)
incognito-dirs
bind/permissions(host)
4. Repository (suspend-only)
Repositories
(manual DI container, lazy)
FavoritesRepository
TabsRepository
HostsRepository
HistoryRepository
DownloadsRepository
2b. Host Seams (view interfaces)
DownloadHost
(bind/unbind + permissions)
WebEngineCallbackHost
AutoUpdateHost
DefaultWebViewDataDirManager
TabsViewHost
WebViewDataDirManager
5. Data (Room)
AppDatabase
FavoritesDao
TabsDao
HostsDao
HistoryDao
DownloadDao